

Composition Processes

Composing with Algorithms
<http://www.bjarni-gunnarsson.net>

"A shift from creating wanted sounds via interactive means, towards creating wanted interactions having audible traces."

Agostino Di Scipio, 'Sound is the interface' (2003)

"The analytical task – given the music, find the rules – is reversed: given the rules, find the music."

G. M. Koenig, quoted by Paul Berg, 'Composing Sound Structures with Rules' (2009)

"The distinctive thing about musical processes is that they determine all the note-to-note (sound-to-sound) details and the over all form simultaneously."

"I want to be able to hear the process happening throughout the sounding music."

Steve Reich, 'Music as a Gradual Process' (1968)

"A system's behaviour is best understood as a trajectory through a space of possibilities structured by attractors, bifurcations, and singularities."

Manuel DeLanda

"Cybernetic music ... is above all that which is no longer 'composed' and rehearsed."

"[The composer] presents his music as an artifice which he constructs and sets in motion, but once he has done this, it is left to move through space, a 'free' music, which ... follows its own internal laws and conditions."

Roland Kayn

"I design systems whose workings I want to explore, not systems that produce a pre-conceived product. ... Technology becomes aesthetically productive when it shifts the possibilities of human perception and collaboration."

Luc Döbereiner

Music that communicates too clearly merely confirms what the listener already knows; music that resists interpretation forces the listener to construct meaning actively.

Herbert Brün, on 'anticommunication'

"Emergence enables forms and behaviors to arise that are not reducible to any single agent."

Dario Sanfilippo

Causality — Di Scipio

“The performance system is no more decomposable into independent parts, it becomes a holon.”

– circular causality

Agostino Di Scipio

“The unit of survival is organism plus environment.”

Cause is a circuit, not an arrow.

Gregory Bateson

Hecker

"I started to think about how I could shift the direction of these compositions ... I was looking into different models of sound perception in psychoacoustics, such as sound being perceived as a 'stream', an 'event', or as an 'auditory object'."

Florian Hecker (in R. Mackay, Sound Out of Line, 2009)

“Composition processes: a process which is unique to every individual when creating music. It usually involves the development of a concept initially, and requires exploring, experimenting and improvising with musical ideas. Ideas are then developed, extended, manipulated and structured into a complete piece of music. The complete piece is often represented in a form appropriate to the genre, such as a traditionally notated score, a lead sheet or a graphic score.”

(Arts Online, Sound Arts: Glossary, 2007)

Composition Methods

An interesting aspect of this definition is how an *initial concept* usually plays an important role.

It implies that the process is carried out in order to *fulfil some intentions*.

It has the purpose of serving the initial concept, but why should the process of making music be *conditioned by a concept*?

Is it actually as useful as the initial inspiration might suggest?

Perhaps it is the *process itself* that is the most important as well as that what is created during the process?

Process

“Although the recipe or instruction set for process production is, or in a certain sense may be, timeless, the process itself must nevertheless be temporal. This means that processes can be said to exist only through their concrete historical manifestations. For processes, to be is to be exemplified [...] this means that the process must exist in time (with its full realization unfolding “in the course of time,” so to speak). As long as it is not concretely realized, we have only a possible and not an actual process.”

(Nicholas Rescher)

Composition Methods

‘Music is traces left by composition.’

(Herbert Brun, Composers and the Computer)

‘Music-making remains an activity revealing its own “creation principle”’

(Horacio Vaggione, Some Ontological Remarks about Music Composition Processes)

‘Musical compositions carry abstract structures’

(Eduardo Miranda, Composing music with computers)

‘Starting from a given sound, many other audibly similar sounds may be developed which however, possess properties different or even excluded from the original sound... No universal tradition of large scale form-building (through these newly accessible sound-relationships) has established itself as a norm.’

(Trevor Wishart, Audible Design)

Material and Form

Material and form have always been related. Microsonic materials and procedures tend to shift the aesthetic focus toward fluid morphologies. The flowing structures that we can create with microsound do not necessarily resemble the usual angular forms of musical architecture. To the contrary, they tend toward liquidic or cloudlike structures.
(***Curtis Roads*** in interview with ***Brigitte Robindore***.)

What is obtained by processes of sonic design, though we may still want to call it sound material, is a formed sound-object. Clearly, as far as timbre is the form of the sound material and sound materials are themselves the object of the composer's knowledge-level strategies, this perspective on composition confounds the classical distinction between material and form.

(***Di Scipio*** - Micro-time Sonic Design)

Compositional Process

A compositional process can be seen as a ***sequence of steps*** which might or might not be iterated in various ways in order to produce a musical result.

Some of these steps might be ***automated*** or carried out by a software program which requires a certain kind of ***clarification*** of what happens in each of the steps.

A compositional process can also be understood as ***navigating a multidimensional space*** of possibilities where one designs the path through potential areas.

Figuring out how these possibilities exist can help in ***designing programs*** that ***explore*** them.

Models

The computer is an ideal tool for *inspecting different aspects* at multiple stages of the creation process.

The study of what happens during the *evolution of a musical composition* made with a computer can give birth to new methods and ways of making music.

This can lead to what Agostino Di Scipio calls '*model of compositional design*'. His model represents a holistic view of the composer's creation process and shows *possible relationships* between various elements that are created during the process and their link to the composition which is its product.

Processes (C.Roads)

1. *Creation of sound source material* (playful)
2. *The important phase of classifying and editing the source material* (types and time-scales).
3. *Make the macroform puzzle.* “It is as if each piece in the puzzle is a sound object with a potentially unique morphology. As I assemble the puzzle, certain objects appear to be natural matches: they fit in sequence or in parallel.”

Processes (R.Chartier)

“My work is really a process of removal. Sometimes a piece will be based on one [looped] sound with things layered over it, and then eventually I will take the original linking element out. So it’s this ghost element that’s not really there. That absence continues to exert a pressure on the work, lurking as an implied presence in the tracks and compounding the obscure impression that something is missing.”

The **organizing principle** is **blurred** by removing important initial material.

Highlights **later choices** as the process unfolds.

Processes (B.Gunter)

“The way I usually start work is with some material that I've treated to a certain degree. That's the starting point. One thing leads to a huge field of possibilities. I add a second thing and the field shrinks, immediately. You have a relationship between the two and there are consequences. When I have that I may repeat one or introduce a new one, and that also has its own consequences. Some people say composing is creating problems and then solving them, but for me it's more like creating possibilities and choosing between them.”

Each **choice that is made** redefines the set of other **possible choices** available at any given point in time.

This unfolding is sequential in nature and could be used to create a **map** of the whole compositional process. Perhaps by doing so its exploration could be repeated with different navigational strategies?

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Processes

It is interesting to think about what happens during the **course of creation**.

The computer is an ideal tool for inspecting different aspects at multiple stages of the **creation process**.

“Composers are concerned with the creation of musical situations emerging concretely out of a critical interaction with their materials, including their algorithms. This task cannot be exhausted by a linear (a priori, non-interactive) problem-solving approach. Interaction is here matching an important feature of musical composition processes, giving room for the emergence of irreducible situations through non-linear interaction.(Vaggione, 2001, p. 54)”

Processes

Intensions should lead to methods that help one *fulfill those intentions*. Once specific methods have been conceived it is interesting to think about their properties and how they can be generalized or used in a different context.

Designing composition software offers *possibilities* to explore ideas regarding *compositional processes*. This could mean applying ideas of organization to the creation of material, manipulating the *relationship between elements forming a process* or *changing values* used in a process progression.

Change

Change

Processes in computer music tend to handle well ***continuity*** and ***repetition***.

An important aspect when dealing with such processes (stochastic or deterministic) is to define ***how they change***.

The most obvious considerations are how a process ***start***, how it ***stops***, how its ***characteristics develop*** and how it ***relates*** to other processes.

These aspects are often highly connected to the ***nature*** of a particular process. This nature can be part of how compositional ***material is defined*** or thought about. We can start with two broad categories, ***textures*** and ***sequences***.

Sequences

A **sequence** is an ordered list of objects. Unlike a set, order matters, and exactly the same elements can appear multiple times at different positions in the sequence.

Perceptible patterns rather than statistical 'clouds'

(James Harley, on the use of percussion in Xenakis Orient-Occident.)

All sequences have two general properties. They can define both a **field** and an **order**. [...] Sequences are also characterized by order properties. In existing musical languages, certain sequences of notes will commonplace others exceedingly unlikely.

'Permutation of a finite set of elements in fact provides a way of establishing relationships among element-groupings. Order can be applied to many different properties and can be used to focus the listener's attention on the ordered property.'

(Trevor Wishart, Audible Design)

Textures

‘Texture [...] is concerned with internal behavior patterning, energy directed inwards or re-injected, self-propagating; once instigated it is seemingly left to its own devices; instead of being provoked to act it merely continues behaving.’
(**Denis Smalley**, Spectromorphology and Structuring Processes)

‘Textural perception [...] only takes over when the succession of events is both random and dense, so we have no perceptual bearings for assigning sequential properties to the sound stream. [...] Texture differs from Continuum in that we retain a sense that the sound event is composed of many discrete events.’

(**Trevor Wishart**, Audible Design)

Evolution

'In general, any parameter we give to a process (pitch, speed, loudness, filter centre, density etc.) should be replaceable by a time varying quantity and this quantity should be continuously variable over the shortest time-scales.'
(**Trevor Wishart**, Audible Design.)

'The metaphors of motion and growth are appropriate ways of considering a time-based art like electro- acoustic music. Traditional concepts of rhythm are inadequate to describe the often dramatic contours of electroacoustic gesture and the internal motion of texture which are expressed through a great variety of spectromorphologies. [...] Motion and growth have directional tendencies which lead us to expect possible outcomes, and they are helpful guides in attributing structural functions.'
(**Denis Smalley**, Spectromorphology: explaining sound-shapes)

Change

“The main thing is: how to change. This is a matter of music, of knowledge, of the universe. Everywhere you feel the changes. The plants are changing, maybe not so fast as the human mind. They’re changing slowly, as the particles do. Probably these particles are changing in the universe on a much larger scale of time. We know at least through astrophysics today that some of them are really mid-life, like the heavy ones. They did not exist at the beginning, and the lighter ones did not exist at the very beginning. So if even the matter itself is changing, everything is changing.”

(Iannis Xenakis)

Transitions

What are the **characteristics** of a decision making regarding a movement from one moment to the next?

Are there maybe a number of **parameters** that define your transition?

Could these be **composed** differently or in a **systematic way** so they and their different characteristics could be explored?

How does one **start** and **stop** processes? Could such a change be developed by their behaviour?

Transitions

Possible ***attributes*** to consider:

Smooth / abrupt

Continuous / discontinuous

From order to disorder

Edges one encounters

Explosions

From clear to unclear and back again.

Separated by silences

Partial (certain qualities transition, others remain)

Transitions

More ***possible attributes*** to consider regarding a transition:

Relationships to overall form and other sections

Rate of change

Amount of activity

Shift of mood

Level of surprise

Transition attack

Temporal Junctions

Curtis Roads (2015) defines musical junctions, meeting points of processes where three possible cases are typical to happen:

- *Continuation* (no change, stasis)
- *Transition* (moderate change)
- *Juxtaposition* (rapid change)

Temporal Junctions

"Continuations include sustaining, repeating, and time dilation processes. A continuation produces different results depending on the structural level on which the composer applies it. On the level of macroform, continuation results in a single movement without distinct sections. An example of a continuous mesostructure would be the prolongation of a tonal center containing many sub-phrases. Continuous phrases feature long and flowing gestures, rather than short and jagged patterns. [...] In contrast, a transition is a sequential change, moving either smoothly or stepwise from one state to another. The possibilities in electronic music include processes such as coalescence, disintegration, and mutation. In electronic music, a transition can involve a granular mutation from one sound to another over time, either continuously or in successive steps. [...] Abrupt juxtapositions articulate mesostructural boundaries. On the macroform level, a typical juxtaposed structure is a set of isolated movements, each distinct from the other. Within a movement, juxtaposed mesostructure appears as clearly demarcated sections. Within a section, juxtaposition interposes distinctly different phrases. Juxtaposition within a phrase occurs if it is constructed out of wildly different sound objects." **(Curtis Roads, 2015)**

Causality

So why and how do events **occur**? What brings one thing **forward** instead of another one?

Processes can be **instantiated** by external forces, interactions or even by other processes.

Sound processes have three important causal features, how they **start**, how they **stop** and how they **relate**.

“Causality, where one event seems to cause the onset of a successor, or alter a concurrent event in some way, is an important feature of acousmatic behavior.” (Smalley, 1997, p. 188).

Causality

Process causality can be **external** and manifested by interaction or manual interventions.

There also exists **internal** process causality, where the concern is with the behaviour of process interiors, usually on a smaller scale. A sound process can thus be started and left alone to expose its internal patterning.

When combining processes on different structural levels, it is useful to speak of **lower-level** and **higher-level** causality. The former is based on what occurs within an event or local section, the latter applies to higher level structures or totalities.

Causality

A composition could for example be made of a handful of formal, higher-level ***structural logics***.

On the lower-levels however, events could happen due to different ***causal functions***.

In this way related processes could be instantiating each other, ***cross-influencing*** their behaviour or characteristics, while at the same time being guided by ***global laws*** determined by a higher-level causality.

Tensions created by the differences or similarities of causal functioning can be interesting for musical purposes and evoke causal relations on several time-scales.

Predictability

Operational levels, where one fixes the window of analysis, are an essential part of nesting processes and causal mechanisms.

Another important aspect is **predictability**. How events succeed each other is part of how their causality is perceived.

How a **succession process** takes place is part of **directionality** and **musical progression**.

“This predictive connection of present and future is held to be the essential feature of causality. Such expectations, whether realized or not, constitute in our imagination goals toward which the music is directed.”
(**Hasty**, 1986, p. 62).

Associations

“The temporality of inner thought processes, is often not linear. Our minds can follow but one branch of the tree of associations; we must return later if we wish to explore another branch. We constantly project fantasies, hopes, and fears into the future; we recall and juxtapose more- and less-remote pasts; we turn our attention from one thought chain to another, often without apparent reason. The temporality of the mind is seemingly irrational.”

(Jonathan Kramer)

Succession

“An observer of an architectural construction chooses a ,”path” with which he goes about the space. This path is built of the sequence of his movements and creates a linear process, a succession of events. Each event is relative to the preceding one (moving from a closed to an open space, from light into the dark etc.), and affects future responses and choices. The observer of a building weaves his path by combining choice with restrictions. He may choose to stop, walk faster or slower, look around, reverse his position in space, but his freedom is confined to certain obstacles on his way, corridors, staircases and points of view that affect his perception.”

(G. Hanoch-Roe)

Emergence

“A musical composition constitutes an emergence because it is not reducible to its parts or elements or even to the operations that were realized during its composition. Therefore we can neglect here the word << Causality >>. [...] The musical work will constitute itself as an emergence (a singularity), and not as a simple globality, since it manifests itself not as that which constitutes the product from the parts of a whole, or even as the epiphenomenon of its components, but as simply as that what it is, without any reduction.”

(Horacio Vaggione)

States

Time

The english physicist Julian Barbour has a theory that ***time does not exist.***

All there exists is a great number of '***nows***'. Time is more of an illusion that we create ourselves by interpolating between them.

This could be relevant in thinking of music as consisting of different ***states*** or nows that we navigate between.

If time is an illusion created by ourselves between the passage of these 'nows' then perhaps music is very much part of this illusion.

Snapshots

“The world is to be understood not as matter/mass moving in a framework of space and time, but of more fundamental snapshot-like entities that momentarily fuse space and matter into single possible arrangements or configurations of the entire universe. Such configurations, which can be fabulously rich and complex considering the vastness of the universe, are the ultimate "things" of reality.”

(Stuart Hameroff)

Networks

“The entities would appear, as in a snapshot, reunited in a dense network of non temporal contiguities, uninterrupted, extending through the entire universe. I said, in a snapshot. That is to say that in the snapshot, the spatial relations of the entities, the forms that their contiguities assume, the structures, are outside time (hors-temps). The flux of time does essentially not intervene in any way. That is exactly what happens with the traces that the phenomenal entities have left in our memory. Their geographical map is outside time.”

(Iannis Xenakis)

Parameters

Parameters can be ***simple*** one-to-one or ***complex*** with a one-to-many behavior.

States of an event object can be set into relationship with the ***parameter states*** of another one.

Inverse, ***Difference*** and ***relative*** mappings are possible.

Specific ***chains of behavior*** control can be created via these relations.

Snapshots

Reference events that remain inscribed somewhere.

Xenakis holds that the **separation** between events is what give creates time since if they ware absolutely smooth time would find itself abolished.

Memory is a spatial translation of the temporal (causal) **chains**. How can we later make a reference to these points in memory and does the temporal order there have to be the same ?

For Xenakis and Barbour the snapshots are not **in time**. They exist outside time and Xenakis holds that this is also true for the traces experienced events have left in our memories.

Outside Time

In the article **ON TIME** Xenakis goes through five steps of how temporal structures are placed outside time.

1. We perceive **separable events**.
2. These events can be assimilated to **points of reference** in the flux of time, but are stored as **outside time** by their traces in our memory.
3. Comparing these events we can assign them **distances, intervals, durations**.
4. We can **repeat, link** these together in a chain
5. There are two possible **orientations in iteration**, one by an **accumulation** of steps, the other by a **deaccumulation**

Helena Gough - Spores (2010)



```

(
  NF(\iop, {|freq=78, mul=1.0, add=0.0|
    var noise = LFNoise1.ar(0.001).range(freq, freq + (freq * 0.1));
    var osc = SinOsc.ar([noise, noise * 1.04, noise * 1.02, noise * 1.08],0,0.2);
    var out = DFm1.ar(osc,freq*4,SinOsc.kr(0.01).range(0.92,1.05),1,0,0.005,0.7);
    HPF.ar(out, 40)
  }).play;
)

(
  NF(\dsc, {|freq = 1080|
    HPF.ar(
      BBandStop.ar(Saw.ar(LFNoise1.ar([19,12]).range(freq,freq*2), 0.2).excess(
        SinOsc.ar( [freq + 6, freq + 4, freq + 2, freq + 8])),
        LFNoise1.ar([12,14,10]).range(100,900),
        SinOsc.ar(20).range(9,11)
      ), 80)
    ).play;
)

var <>pindex, <>cindex;

initialize {
  if(pindex.isNil, { pindex = 1000 });
  if(cindex.isNil, { cindex = 2000 });
}

clearProcessSlots {
  pindex = 1000;
  (this.pindex - 1000).do{|i| this[this.pindex+i] = nil; }
}

clearOrInit {|clear=true|
  if(clear == true, { this.clearProcessSlots() }, { this.initialize() });
}

transform {|process, index|
  if(index.isNil && pindex.isNil, {
    this.initialize();
  });

  pindex = pindex + 1;
  this[pindex] = \filter -> process;
}

control {|process, index|
  var i = index;

  if(i.isNil, {
    this.initialize();
    cindex = cindex + 1;
    i = cindex;
  });

  this[i] = \pset -> process;
}

(
  NF(\depfm, {|freqMin=5, freqMax=20, mul=20, add=80, rate=0.5, modFreq=2100, index=0.3, amp=0.2|
    var trig, seq, freq;
    trig = Dust.kr(rate);
    seq = Diwhite(freqMin, freqMax, inf).midicps;
    freq = Demand.kr(trig, 0, seq);
    HPF.ar(PMOsc.ar(LFCub.kr([freq, freq/2, freq/3, freq/4], 0, mul, add),
      LFNoise1.ar(0.3).range(modFreq,modFreq*2), index) * amp, 50)
  }).play;
)

```

Exercises

Exercises

1. Develop a method for selecting between various pre-configured Pbinds so that they can be played in different order in time.
2. Create a sequencing process that plays at least two different sections. Each of the section should contain parameters that control its behaviour and the order of sequences should be flexible.
3. Create three different snapshots of synthesiser configuration using the SynthDef designer tool. Later, implement at least two different patterns playing these synthdefs.
4. Implement at least one transition from a certain material to another set